

EfficientZSR: A Lightweight Generalized Zero-Shot Recognition Framework

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Abstract—Within the realm of deep learning, Zero-Shot Learning (ZSL) presents a significant challenge, particularly in resource-constrained environments where efficient model inference is crucial. To tackle this problem, we present EfficientZSR, a lightweight generalized zero-shot recognition framework that integrates the Vision Transformer (ViT) as the teacher model and EfficientNet as the student model. Through the application of knowledge distillation techniques, EfficientZSR effectively inherits the powerful feature extraction capabilities of ViT while maintaining a lightweight structure. We performed evaluations across three standard datasets. (AWA2, CUB, and SUN) and achieved outstanding results across all three, further validating the effectiveness and broad applicability of EfficientZSR.

Keywords—Zero-Shot Learning, Vision Transformer, Efficientnet, DIST.

I. INTRODUCTION

With advancements in computer vision, deep learning models excel at feature extraction and classification but require large labeled datasets and can only recognize categories seen during training. This limitation led to the development of Zero-Shot Learning (ZSL) [1], which recognizes unseen categories by leveraging semantic information from seen categories, such as attributes or word embeddings.

Although Zero-Shot Learning (ZSL) holds significant theoretical potential, it still faces numerous challenges in practical applications. Traditional ZSL methods typically include only unseen category samples in the test set. Nevertheless, in practical real-world applications, the number of seen category samples often exceeds that of unseen ones, necessitating the recognition of both seen and unseen categories, a task known as Generalized Zero-Shot Learning (GZSL) [2].

Despite the substantial promise of GZSL in knowledge transfer and the recognition of unseen categories, most existing GZSL methods rely on complex and resource-intensive models, such as f-CLSWGAN and CVAE-SZL [3, 4]. To resolve this problem, we present a lightweight generalized zero-shot recognition framework, EfficientZSR, based on knowledge distillation. This framework utilizes the Vision Transformer (ViT) [5] as the teacher model and EfficientNet-B0 [6] as the student model, with knowledge transfer via DIST [7]. Compared to traditional GZSL approaches, EfficientZSR reduces resource consumption while maintaining high performance. We validated it on CUB, AWA2, and SUN datasets, showing its superiority.

II. METHODOLOGY

The EfficientZSR model utilizes the ViT-L/16 as the teacher network and EfficientNet-B0 as the student network to extract features from images of seen classes. The structure of the EfficientZSR model is shown in Fig. 1.

During training, the images are first passed through both the ViT and EfficientNet-B0 models to extract features. The features generated by EfficientNet-B0 are aligned with those generated by ViT through a distillation loss. Simultaneously, EfficientNet-B0 combines these features with semantic attributes to generate predicted labels, which are compared with the ground truth labels using a cross-entropy classification loss. The distillation loss and classification loss are combined to form the total loss, guiding the optimization and training of the EfficientNet-B0 model.

During testing, the distilled student network EfficientNet-B0 is used to extract features from images of unseen classes. The predicted labels are then obtained by calculating the cosine similarity between these features and the image attributes [8], and the accuracy is computed by comparing the predicted labels with the ground truth labels.

A. Teacher Model

In the EfficientZSR framework, we selected the Vision Transformer (ViT) [5] as the teacher model to guide the learning process of the student model, EfficientNet. The core concept of ViT is to divide the image into fixed-size patches, which are then linearly embedded and fed into a Transformer encoder. Unlike traditional Convolutional Neural Networks (CNNs), ViT does not rely on local receptive fields and translation invariance but directly processes global information from image patches.

During the training process, the set of seen classes $S_{seen} = \{x, y \mid x \in X^{seen}, y \in Y^{seen}\}$ is used as input images, where x represents an RGB image, and y is a class-level attribute vector annotated with M distinct attributes. First, a 224×224 image $x \in \mathbb{R}^{H \times W \times C}$ is divided into patches of size $P \times P$. Each patch is then flattened into a sequence $x_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$, where $N = HW/P^2$ is the number of patches. These flattened patches are projected into a dimensional space D through a linear mapping. To preserve positional information, a learnable positional embedding vector x_{class} is added to each patch. These positional embeddings, along with the patch embeddings, are fed into the Transformer encoder. The encoder consists of alternating layers of Multi-Head Self-

Attention (MSA) and Multi-Layer Perceptron (MLP) blocks. Layer normalization is applied before each block, and residual connections are employed after each block.

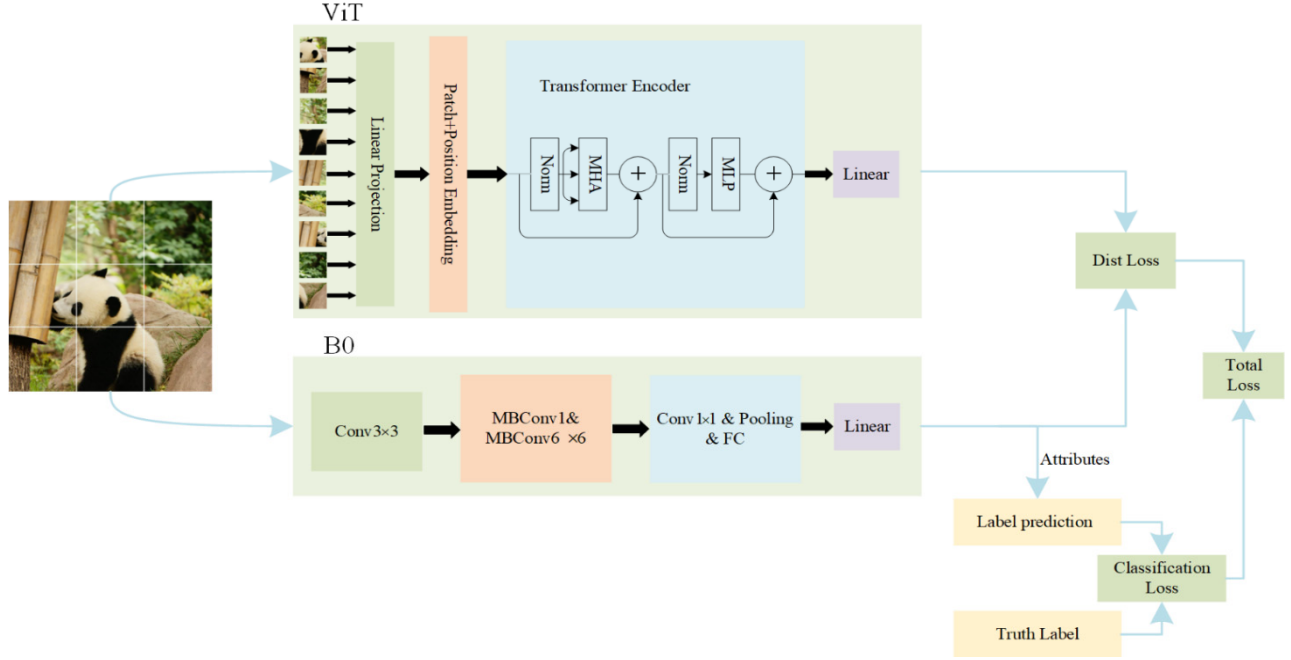


Fig. 1. The architecture of EfficientZSR.

Multi-Head Attention (MHA) serves as a crucial element of the Vision Transformer (ViT). MHA performs self-attention operations across multiple subspaces, capturing the dependencies between different parts of the input image. Specifically, MHA first projects the input query (Q), keys (K), and values (V) matrices into multiple subspaces where self-attention is computed independently within each subspace:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Here, d_k represents the dimension of the keys vector, and $\frac{1}{\sqrt{d_k}}$ is a scaling factor to prevent the dot products from becoming excessively large.

The outputs from each subspace are then concatenated to form the final output. This mechanism enables the model to concentrate on distinct features of the image within different subspaces, thereby enhancing the ability to grasp both local details and global context.

B. Student Model

In the EfficientZSR framework, the student model is implemented using EfficientNet-B0. The core idea of EfficientNet is to improve performance by uniformly scaling all dimensions of the model—depth, width, and resolution—instead of merely increasing a single dimension. By employing a compound scaling strategy, EfficientNet significantly reduces computational complexity and the number of parameters while

maintaining high accuracy. The key concept of the compound scaling method is to simultaneously adjust the network's depth, width, and resolution using a unified scaling coefficient ϕ . The formula

is as follows:

$$\begin{aligned} \text{depth} : d &= \alpha^\phi \\ \text{width} : w &= \beta^\phi \\ \text{resolution} : r &= \gamma^\phi \\ \text{s.t. } \alpha \cdot \beta^2 \cdot \gamma^2 &\approx 2 \\ \alpha \geq 1, \beta \geq 1, \gamma \geq 1 \end{aligned} \quad (2)$$

Here, α , β and γ are constants identified via grid search and represent the scaling coefficients used to control resource usage.

By fixing α , β and γ , and adjusting the value of ϕ , the network can be expanded, resulting in models ranging from EfficientNet-B1 to B7. The baseline network, EfficientNet-B0, is designed through Neural Architecture Search (NAS) and primarily utilizes Mobile Inverted Bottleneck Convolution (MBConv) layers [9, 10], combined with Squeeze-and-Excitation (SE) optimization [11]. As the foundational model in the EfficientNet series, EfficientNet-B0 strikes an exceptional balance between performance and computational cost, making it a representative example of lightweight models.

C. the Distillation Loss Function

In the EfficientZSR framework, the distillation loss function—DIST [7], is a critical component that effectively facilitates knowledge transfer by capturing both inter-class and intra-class relationships between the teacher and student models. Traditional knowledge distillation methods typically rely on Kullback-Leibler (KL) divergence to minimize the output distribution differences between the teacher and student models [12]. However, when the teacher model is significantly more powerful than the student model, this direct output matching can lead to severe learning difficulties and may even degrade the performance of the student model. This is because the prediction distributions generated by the teacher model are often more refined and complex, making it challenging for the student model to accurately replicate these outputs.

DIST alleviates the aforementioned issue by introducing a correlation-based loss function. The core idea is to preserve the inter-class and intra-class relationships between the teacher and student models rather than strictly matching their outputs. In the DIST (Distillation from A Stronger Teacher) method, the inter-class loss is realized by maximizing the Pearson Correlation Coefficient between the prediction distributions of the teacher and student models. The formula is as follows:

$$L_{inter} = \frac{1}{B} \sum_{i=1}^B (1 - \rho_p(Y_i^{(s)}, Y_i^{(t)})) \quad (3)$$

Here, $Y_i^{(s)}$ and $Y_i^{(t)}$ represent the prediction distributions of the student and teacher models, respectively, and ρ_p denotes the Pearson Correlation Coefficient.

The intra-class loss is realized by maximizing the correlation between the prediction distributions of different samples within each class. The formula is as follows:

$$L_{intra} = \frac{1}{C} \sum_{j=1}^C (1 - \rho_p(Y_{:,j}^{(s)}, Y_{:,j}^{(t)})) \quad (4)$$

Here, $Y_i^{(s)}$ and $Y_i^{(t)}$ represent the prediction distributions of the student and teacher models for class j , respectively, and ρ_p denotes the Pearson Correlation Coefficient.

The DIST loss function is defined as follows:

$$\mathcal{L}_{tr} = \kappa \mathcal{L}_{cls} + \mu \mathcal{L}_{inter} + \nu \mathcal{L}_{intra} \quad (5)$$

Here, $\mathcal{L}_{cls}$ represents the cross-entropy loss between the student model's predictions and the ground truth labels, $\mathcal{L}_{inter}$ represents the inter-class relationship loss, and $\mathcal{L}_{intra}$ represents the intra-class relationship loss.

III. EXPERIMENTS

A. Datasets

To verify the efficacy and lightweight characteristics of the presented EfficientZSR framework in zero-shot learning tasks,

we performed a series of tests using one coarse-grained dataset and two fine-grained datasets: Caltech-UCSD Birds-200-2011 (CUB) [13], Animals with Attributes 2 (AWA2) [14], and Scene Understanding (SUN) [15]. As shown in Table 1, the CUB dataset contains 11,788 images of 200 bird species, with 150 species used as the training set (seen classes) and the remaining 50 species as the test set (unseen classes). The CUB dataset provides 312 attribute labels for each class to describe the various characteristics of the birds. The AWA2 dataset consists of 37,322 images of 50 animal species, with 40 species used for training and 10 species for testing. The AWA2 dataset offers 85 attribute labels to describe different features of the animals. The SUN dataset encompasses 14,340 images across 717 distinct scenes, with 645 scenes used for training and 72 for testing. This dataset includes 102 attribute labels per class, used to describe the characteristics of the scenes.

Table 1 Experimental Datasets.

Datasets	Granularity	Attributes	Classes (S+U)	Images
SUN	fine	102	717(645+72)	14340
CUB	fine	312	200(150+50)	11788
AWA2	coarse	85	50(40+10)	37322

B. Implementation Details

In our implementation, the Vision Transformer (ViT) served as the teacher model with fixed pre-trained weights, while only the student model, EfficientNet-B0, underwent parameter optimization. The Adam optimizer was employed for EfficientNet with a base learning rate of 1e-5 and a reduced learning rate of 1e-3 for the linear layers. To address the sensitivity to overfitting, a weight decay strategy was implemented with a base coefficient of 1e-4 and 1e-5 for the linear layers. We also utilized a Cosine Annealing Learning Rate (LR) Scheduler with a maximum of 30 epochs to fine-tune the learning rate dynamically [16]. To enhance the model's performance on both seen and unseen classes, the Calibrated Stacking technique was applied [2], with a sensitivity analysis on the calibration factor δ . This factor was adjusted based on the dataset: 0.9 for AWA2 and CUB, and 0.5 for SUN, demonstrating our model's adaptability to different scenarios.

C. Evaluation

In the context of generalized zero-shot image classification, we computed the top-1 accuracy for both seen and unseen classes and used the harmonic mean H as the evaluation metric. The value of H reaches its maximum when the accuracy of seen classes (S) and unseen classes (U) is balanced. The formula for calculating H is as follows:

$$H = \frac{2 \times S \times U}{S + U} \quad (6)$$

D. The Comparison of Results

Experiments were conducted on the aforementioned datasets, and this method was compared with several other state-of-the-art GZSL methods, as shown in Table 2, where the highest values are highlighted in bold and italics.

The model, EfficientZSR, achieved the highest harmonic mean of 72.7% on the AWA2 dataset, with an accuracy of 90.3% on seen classes, ranking second. On the CUB dataset, EfficientZSR achieved the second highest accuracies for both seen and unseen classes, recording 68.9% and 60.0% respectively. Furthermore, it reached the highest harmonic mean of 64.1%. In the SUN dataset, EfficientZSR also exhibited stable performance with a harmonic mean of 44.7%. Although its accuracy on seen classes was slightly lower, its performance on unseen classes compensated for this, maintaining EfficientZSR's competitive overall performance.

E. Hyperparameter Analysis

To address the sensitivity of EfficientZSR's performance to key parameters, we conducted a comprehensive analysis focusing on the distillation loss weights. The hyperparameters

Table 2 Gzsl Result.

Models	AWA2			CUB			SUN		
	<i>S</i>	<i>U</i>	<i>H</i>	<i>S</i>	<i>U</i>	<i>H</i>	<i>S</i>	<i>U</i>	<i>H</i>
DAP[17]	84.7	0.0	0.0	67.9	1.7	3.3	25.1	4.2	7.2
DeViSE[18]	74.7	17.1	27.8	53.0	23.8	32.8	30.5	14.7	19.8
ConSE[19]	90.6	0.5	1.0	72.2	1.6	3.1	39.9	6.8	11.6
SSE[20]	82.5	8.1	14.8	46.9	8.5	14.4	36.4	2.1	4.0
SJE[21]	73.9	8.0	14.4	59.2	23.5	33.6	30.5	14.7	19.8
ESZSL[22]	77.8	5.9	11.0	63.8	12.6	21.0	27.9	11.0	15.8
SRSA[23]	59.6	38.1	46.5	55.6	27.5	36.8	37.9	25.3	30.3
DAZLE[24]	75.7	60.3	67.1	59.6	56.7	58.1	24.3	52.3	33.2
LMILR[25]	57.5	83.9	68.2	52.4	57.9	55.0	47.9	36.4	41.4
f-CLSWGAN[3]	52.1	68.9	59.4	43.7	57.7	49.7	42.6	36.6	39.4
TF-VAEGAN[4]	59.8	75.1	66.6	52.8	64.7	58.1	45.6	40.7	43.0
SE-GZSL[26]	80.7	59.9	68.8	60.3	53.1	56.4	40.7	45.8	43.1
Our model	90.3	60.8	72.7	68.9	60.0	64.1	36.9	56.7	44.7

$\mathcal{K}$, μ , and ν are crucial for balancing the classification loss, inter-class loss, and intra-class loss, respectively. Through

a systematic grid search, we optimized these parameters for each dataset. For AWA2, CUB, and SUN, the selected values were (0.9, 0.01, 0.01), (0.1, 0.1, 0.1), and (0.9, 0.03, 0.03), respectively. These settings were chosen to reduce the influence of parameter variations on model performance, ensuring robustness across different datasets. Our analysis, as depicted in Fig. 2, highlights the harmonic mean values associated with various hyperparameter combinations, demonstrating the model's sensitivity to these critical settings.

F. Distillation Performance Comparison

Tables 3 compare the differences in parameter count and performance between the teacher model (ViT) and the student model (EfficientZSR) across the three datasets.

The results indicate that EfficientZSR not only achieves excellent performance in generalized zero-shot learning (GZSL) tasks but also demonstrates significant advantages in lightweight design. The ViT teacher model contains 304.3M parameters, whereas the EfficientNet-B0 student model has only 5.3M parameters, approximately 1/57th of the ViT's parameter count. This means that EfficientNetZSR substantially reduces the demand for computational and storage resources, offering feasibility for deployment in a wide range of application scenarios, especially in resource-constrained edge devices and embedded systems.

Despite the drastic reduction in parameter count, EfficientNetZSR still performs remarkably across multiple benchmark datasets. In the AWA2 dataset, the harmonic mean (*H*) of EfficientNetZSR is 72.7%, which is 2.6% higher than the ViT's 70.1%. The result highlights that EfficientNetZSR not only maintains competitive performance but even surpasses its teacher model. In the CUB dataset, EfficientNetZSR's harmonic mean is 64.1%, lower than ViT's 69.0%, but it remains competitive in accuracy for unseen classes (60.0% compared to 68.0%). In the SUN dataset, EfficientNetZSR performs less than ViT, with a harmonic mean of 44.7%, because of the complexity of the SUN dataset scenarios. And the poor performance of the student model may be due to the limitations in the learning capacity of the teacher model. These results indicate that the inter-class and intra-class information provided by ViT through distillation is well-preserved in EfficientNet-B0, allowing the student model to maintain high levels of classification accuracy while significantly reducing the model size. So EfficientNetZSR maintains strong generalization capability even in complex classification tasks.

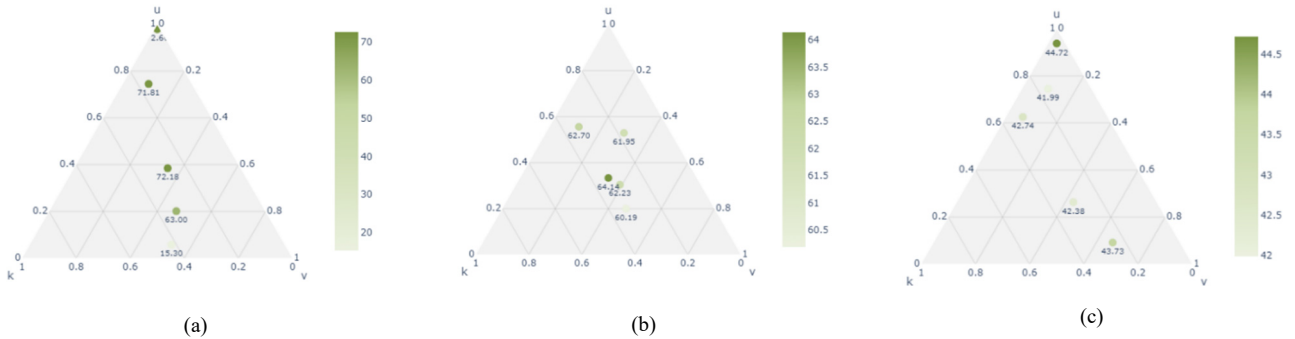


Fig. 2. (a) is the cone chart of AWA2, (b) is the cone chart of CUB and (c) is the cone chart of SUN.

Table 3 Performance Comparison Between ViT And Efficientzsr.

Models	Params	AWA2			CUB			SUN		
		<i>S</i>	<i>U</i>	<i>H</i>	<i>S</i>	<i>U</i>	<i>H</i>	<i>S</i>	<i>U</i>	<i>H</i>
ViT	304.3M	91.8	56.7	70.1	69.9	68.0	69.0	50.8	48.0	49.3
EfficientZSR	5.3M	90.3	60.8	72.7	68.9	60.0	64.1	36.9	56.7	44.7

IV. CONCLUSION

We introduce EfficientZSR, a compact zero-shot recognition framework that leverages knowledge distillation to integrate ViT's feature extraction with EfficientNet's efficiency. Utilizing the DIST loss function, EfficientZSR boosts performance on zero-shot tasks and maintains a lightweight structure, with a parameter count roughly 1/57th that of ViT. This framework is well-suited for applications like public safety surveillance, smartphone image classification, autonomous driving object detection, and smart home tech, as demonstrated by its strong accuracy on AWA2, CUB, and SUN datasets.

Looking ahead, there is potential to further improve EfficientZSR by refining its model structure and distillation techniques. This would enhance its capacity for real-time processing, managing large datasets, and adapting across various tasks. Moreover, to bolster its applicability in dynamic environments, ongoing work will concentrate on enhancing its flexibility to cope with emerging, unseen categories and fluctuating data distributions. This will render EfficientZSR more robust in open-world scenarios, where adaptability is paramount. Additionally, the framework's potential in handling real-world zero-shot learning problems, such as image recognition in low-resource settings and dynamic environments, will be investigated more deeply.

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